MIL-C-29213A(YD)
15 December 1987
SUPERSEDING
MIL-C-29213(YD)
17 March 1982

MILITARY SPECIFICATION

CONCRETE PLACING BOOM, TRUCK MOUNTED, 6 X 4, DIESEL-ENGINE-DRIVEN, COMMERCIAL

This specification is approved for use by the Naval Facilities Engineering Command (YD), Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of commercial, diesel engine driven, 6 X 4, truck mounted, concrete placing boom with an option for a concrete pump to be furnished. Trucks procured under this specification are commercial and shall be warranted by the manufacturer as specified in procurement documents.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified (see 6.2), the following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

FEDERAL

W-B-131 - Battery, Storage: (Vehicular, Ignition, Lighting and Starting).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3895

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:17Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-1223 - Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking, Data Plate Standards.

(Copies of specifications, standards, handbooks, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Vehicle Safety Standards and Regulations.
Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines: Certification and Test Procedures.
Interstate Motor Carrier Noise Emission Standards.
Motor Vehicle Air Pollution Standards.

(Application for copies should be addressed to the Public Affairs Office, Environmental Protection Agency, Rockville, MD 20852; or CFR, Title 40 should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:17Z
Check the source to verify that this is the current version before use.

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

STATE OF CALIFORNIA

California Vehicle Code.

(Application for copies should be addressed to the Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The truck mounted concrete placing boom which consists essentially of a 6 X 4 chassis, diesel engine driven, minimum 46,000 pound (lb) gross vehicle weight (gvw) truck complete with hydraulically controlled articulating type concrete placing boom, hydraulic rotation assembly, hydraulic outriggers, and an option for a concrete pump to be furnished. The mounted equipment and truck shall herein be referred to as the unit.

3.2 First article. When specified (see 6.2), in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1, 6.2, and 6.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:17Z
Check the source to verify that this is the current version before use.

3.3 Standard commercial product. The unit shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the unit being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.6 Maintainability. The unit design, together with component and accessory location and installation, shall permit ready accessibility to all items requiring periodic maintenance. Service in the field, which will be accomplished with the use of conventional general-purpose tools associated with equipment of this nature. The replacement and adjustment of components and accessories shall be accomplished with minimum drainage requirements and minimum disturbance to other elements of the unit.

3.7 Transportability. The unit complete with accessories and the boom lowered to transport position, shall be transportable without damage or permanent deformation over unpaved roads and rough terrain at speeds as great as 15 miles per hour (mph), and on highways up to 55 mph. The overall height of the boom in the transport position shall be not more than 13 feet 6 inches above the ground.

3.8 Safety. All rotating or reciprocating parts and all parts which are subject to high operating temperatures, and moving parts which are of such nature or so located as to constitute a hazard to personnel shall be insulated, enclosed, or guarded. Safety devices and safety measures shall not impair the required functioning of the unit. Principal platform walking surfaces shall be of an antiskid type.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:17Z
Check the source to verify that this is the current version before use.

3.9 General design.

3.9.1 DoT Federal Motor Vehicle Safety Standards. The truck and furnished accessories shall comply with all DoT Federal Motor Vehicle Safety Standards in effect on the date of manufacture.

3.9.2 Air pollution control. The truck shall comply with the EPA Regulations governing Control of Air Pollution from New Motor Vehicles and new Motor Vehicle Engines in effect on the date of manufacture. In addition, vehicles destined for California shall comply with State of California regulations governing air pollution control in effect on date of manufacture.

3.9.3 Sound level. The interior sound level shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.94. The truck exterior sound level shall conform to the EPA Interstate Motor Carrier Noise Emission Standards.

3.9.4 Curb weight. Curb Weight shall include weight of chassis and cab, with all attachments, accessories, and equipment; full complement of fuel, lubricants, and coolant.

3.9.5 Gross vehicle weight ratings. GVW shall consist of curb weight, operator (weight computed at 175 lb), and a capacity to provide not less than 46,000 lb.

3.9.6 Weight distribution. Suspension, axle, and tire capacities and ratings shall be not less than 12,000 lb on the front and 34,000 lb on the rear.

3.9.7 Maximum geared speed. Maximum geared speed at engine governed speed shall not be less than 55 mph (-1 percent). Conformance to geared speeds specified shall be determine by calculating in accordance with the following formula:

$$\text{Maximum geared speed (mph)} = \frac{\text{governed speed revolutions per minute}}{\text{total gear reduction X tire factor}}$$

3.9.8 Service brakes. The service brakes shall control and hold the vehicle, when loaded to its specified gross weight, on a 30 percent grade. The service brakes shall stop the vehicle, loaded to specified gvw, within the stopping distance requirements of DoT Federal Motor Carrier Safety Regulation 393.52.

3.9.9 Drain plugs. Drain plugs installed in manual transmission and rear axles shall be of the permanent-magnet type.

3.10 Chassis components.

3.10.1 Engine. Engine furnished shall be the chassis manufacturer's standard, or optional engine for the commercial model truck which meets or exceeds the requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:17Z
Check the source to verify that this is the current version before use.

3.10.1.1 Diesel engine. The vehicle shall be equipped with a liquid cooled, compression ignition, two-stroke, or four-stroke cycle diesel engine, with not less than six cylinders. Engine net horsepower (hp) figures used in performance prediction calculations shall be determine in accordance with SAE J816. A fan clutch to reduce fan speed automatically, when not required for engine cooling, shall be provided.

3.10.1.2 Fuel economy diesel engine. Engines rated at more than 220 net hp shall be the manufacturer's recommended fuel economy version of a conventional (intermittent brake hp rated) engine. The fuel economy engine shall have high torque-rise capabilities and shall be governed to the engine manufacturer's fuel economy (continuous brake hp rated), full-load revolutions per minute (rpm) configuration, but shall not exceed 1950 rpm. The fuel economy configuration shall be attained by alterations and displacement. The fuel economy engine shall be turbocharged and shall have an inlet air after-cooler, if aftercooling is the engine manufacturer's standard practice for the model engine furnished.

3.10.1.3 Oil filter. Full flow type oil filter shall be furnished.

3.10.1.4 Governor. Engine governor shall be furnished and set and sealed to prevent exceeding the engine manufacturer's maximum recommended operating speed.

3.10.1.5 Coolant temperature control. Thermostatic control of engine coolant temperature shall be provided. Control shall include partial thermostatic control of coolant flow through the radiator and thermostatically controlled radiator shutters or complete thermostatic control of all coolant flow through the radiator.

3.10.2 Electrical system. Electrical system shall be in accordance with DoT Federal Motor Carrier Safety Regulations, section 393.27 through 393.31, and 393.33.

3.10.2.1 Starting system. A 12- or 24-volt (V) starting system, with 12-V lighting system, and with not less than 60-ampere (A) alternator, shall be furnished. Engine starting equipment shall include an ether starting system or glow plug. The diesel engine shall start in any temperature above -20 degress Fahrenheit (°F). When an ether priming system is required, it shall be of the measured shot type with reservoir capacity of not less than 12 fluid ounces. Measure shot type ether systems shall be key operated or manually operated from the driver's compartment and shall be inoperative with the engine warm.

3.10.2.2 Lighting. All truck lights, reflectors, and wiring shall be as specified herein and shall be in conformance with DoT Federal Motor Carrier Safety Regulations, section 393.12, 393.13, 393.19, 393.20, and 393.22 through 393.26(D). Lights and reflectors shall not be mounted on vertical surface of rub rails or mounted on vehicle bumpers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:17Z
Check the source to verify that this is the current version before use.

3.10.2.3 Turn signals. Turn signal lamps shall conform to SAE J588. Operating units shall conform to SAE J589, class A, and shall be mounted on the steering column. Truck shall be provided with double-faced front signal units and with single-faced rear signal units installed in accordance with SAE J588, except that turn signals shall not be mounted on engine compartment hood. Turn signal operating units shall be visible and audible flash indicators.

3.10.2.4 Batteries. Each battery shall be 12-V potential. The total reserve capacity rating shall be not less than 320 minutes and the total cold cranking rating at 0°F shall not be less than 1200A, measured in accordance with SAE J537. Unless otherwise specified (see 6.2), batteries shall be of the maintenance-free type having the maintenance-free characteristics listed in W-B-131. When specified (see 6.2), batteries shall be of the dry-charge type, sealed to prevent atmospheric moisture from entering until put into service.

3.10.3 Fuel system. Fuel system shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.65 and 393.67.

3.10.3.1 Air cleaner. Manufacturer's standard air cleaner shall be furnished.

3.10.3.2 Fuel tank(s). Fuel tank(s) shall not be less than 40 gallons total capacity. When more than one tank is furnished, means shall be provided to assure equalized fuel level in both tanks.

3.10.4 Exhaust system. The exhaust system shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.83. Exhaust mufflers, when stacked, shall be furnished with a heat shield. Stacked exhaust system shall be provided with a hinged rain cap.

3.10.5 Transmission. Unless otherwise specified (see 6.2), a manually shifted transmission shall be provided on the truck. Input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. Gear ratios in transmission and axles shall be matched to provide a progressive shifting pattern through the complete range, and shall provide required vehicle performance (see 3.9.7). The transmission shall be provided with two Power Takeoff Openings (pto) conforming to SAE J704. When a fuel economy engine is furnished (see 3.10.1.2), the transmission shall be of the "wide-step" design, with gear ratio steps not less than 30 percent and not more than 40 percent, except low gear or first gear in each range may have a ratio step exceeding 40 percent. The transmission shall have not less than six and not more than nine speeds. The transmission design shall assure maximum vehicle performance in all gears with the engine rpm above the peak torque rpm and in the engine economy rpm range. When a conventional (intermittent brake hp rated) engine is furnished, the transmission shall be the manufacturer's standard design with not less than 10 forward speeds. A multispeed tandem axle, multirange transmission, or an auxiliary transmission shall be provided with conventional engines.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:17Z
Check the source to verify that this is the current version before use.

3.10.5.1 Clutch. Clutch shall be the largest capacity clutch offered for the truck and engine furnished with torque capacity exceeding maximum delivered engine torque.

3.10.5.2 Power takeoff. When a pto is furnished, it shall be of the rated capacity required to operate equipment to be powered. Controls to operate the pto shall be located in the truck cab accessible to the seated driver. A caution plate or decal reading "DO NOT OPERATE VEHICLE AT HIGHWAY SPEEDS WITH POWER TAKEOFF ENGAGED" shall be provided and installed so as to be readily visible to the seated driver.

3.10.5.3 Automatic or semiautomatic transmission. When specified (see 6.2), the truck shall be provided with an automatic or semiautomatic transmission. The input torque capacity of the transmission shall be at least equal to the maximum torque delivered by the engine. The transmission shall include a hydraulic torque converter and not less than four forward gear ratios. Normal driving range selector position shall provide not less than four gear ratios without movement of the selector. The transmission shall be provided with an SAE J704 power takeoff opening.

3.10.6 Drive line components. Drive line components shall be adequate to transmit the maximum delivered torque of the engine, as developed through the maximum gear train reduction.

3.10.7 Frame. Chassis frame shall be manufacturer's heavy-duty main frame or frame with reinforcement extending at least from the rear of the front-suspension, rear hanger bracket to the bogie trunnion mounting bracket. Heavy-duty frame shall have frame rails of greater section modules than the manufacturer's standard, and shall provide structural strength at least equivalent to the reinforced frames specified herein.

3.10.8 Suspension. The truck shall be equipped with a suspension system with components having a rated capacity of at least equal to that specified in 3.9.6. When spring capacity is rated at the spring pads, unsprung weight shall be deducted. Truck shall be equipped with hydraulic, double-acting shock absorbers in the front wheels. Auxiliary rear springs shall be provided.

3.10.9 Axles. Axle ratings shall be at least equal to that specified in 3.9.6. Single reduction rear axles shall be provided.

3.10.9.1 Rear bogie. The rear bogie shall be of the four-wheel type, complete with axles, springs, torque rods and all other necessary parts. The bogie shall be provided with means permitting differential action between the two axles, and a manually or automatically controlled lockout assuring equal power to each rear axle. Manual locking control used shall be located in the truck cab. When the truck is equipped with a high torque-rise fuel economy diesel engine (see 3.10.1.2), it shall have an axle with a numerical ratio of not more than 4 to 1. Axle gear ratios shall provide performance as specified in 3.9.7.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:17Z
Check the source to verify that this is the current version before use.

3.10.10 Wheels, rims, tires, and tubes. Truck shall be equipped with disc type, single front and dual or wide-base rear wheels. Rim and tire rating shall conform to TRA recommendations, for the type and size of tires furnished. Tire and rim sizes shall be the same for all wheels. Ten stud disk type wheels shall be furnished.

3.10.10.1 Tires. Tires shall be bias ply. Tires shall be tube or tubeless type with highway tread. Tires shall be of rated capacity at least equal to that specified in 3.9.6.

3.10.10.2 Inner tubes. For tube tires, inner tubes shall be of heavy-duty type, and shall be of proper size for tires furnished. Tire flaps shall be provided for tube type tires in accordance with TRA recommendations.

3.10.10.3 Spare wheel assembly. A spare wheel with a tire shall be mounted on a carrier and installed in a readily accessible location. The tire shall meet the requirements specified in 3.10.10.1.

3.10.11 Brakes. Brakes shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.40 through 393.40 through 393.42(b), 393.43, and 393.45 through 393.52.

3.10.11.1 Service brakes. Truck shall be equipped with full-air brakes on all wheels. The braking system, complete with all necessary components, shall include:

a. Air compressor, unloader-head-type, engine driven and engine lubricated, air or water cooled, and having a capacity of not less than 12 cubic feet per minute.
b. Air storage reservoir(s) with not less than 3,000 cubic inches total capacity; each tank equipped with drain, and with safety and check valves between compressor and last reservoir tank.
c. Foot control, suspended or treadle type.
d. Air control valves.
e. Air-pressure gage, visible to the driver.
f. Low-air-pressure warning, visible and audible.
g. Service brake stop lamp switch.
h. Automatic moisture ejector.
i. Automatic slack adjusters.

3.10.12 Cab. Manufacturer may furnish any type of his standard or optional full-width cab. Cab doors shall be equipped with locks, operable from inside the cab through mechanical linkage, and with at least the curbside door equipped with external, key-operated lock. Drip rails shall be installed above the doors. Cab shall have upholstered, full-width, adjustable seat and back, or individual adjustable driver's seat, and individual passenger seat. White upholstery shall not be furnished. Safety grips shall be provided on each side of cab to assist personnel to enter or leave cab. Interior lighting shall be provided. Fastenings and two pairs of seat belts shall be installed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:17Z
Check the source to verify that this is the current version before use.

3.10.13 Steering. Power steering shall be furnished.

3.10.14 Windshield wipers and washers. Truck shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multispeed type operated by air or electric motor(s).

3.10.15 Bumper. Unless bumper is an integral part of truck cab, a channel type front bumper shall be provided.

3.10.15.1 Rear end protection. Rear end truck shall be protected in accordance with DoT Federal Motor Carrier Safety Regulations, section 393.86.

3.10.16 Tool stowage. Stowage space of sufficient size to accomodate truck jack, hand tools, tire chains, and emergency reflective triangles shall be furnished for retaining equipment during vehicle operation. Stowage space for these tools may be furnished inside the cab. When stowage space for these tools is located outside of the cab, it shall be waterproof and shall provide for locking.

3.10.17 Heater and defroster. Hot water shall be provided. Heater shall have fresh air intakes. Discharge outlets shall be provided to direct heated air to floor and to defroster louvers. Heater shall be complete with blower and mounted controls convenient to the driver.

3.10.18 Controls and operating mechanism. All controls and operating mechanism shall be located for left-hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exist of operator to and from driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing.

3.10.19 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment by the manufacturer. The following minimum equipment shall be furnished:

a. Key-operated ignition switch.
b. Ammeter, charging indicator, or voltmeter.
c. Fuel gage.
d. Oil pressure gage or indicator.
e. Engine temperature gage or indicator.
f. Speedometer with recording odometer.
g. Tachometer.
h. Ash receptacle.
i. Cigar lighter
j. Dual sunvisors.
k. Driver's compartment ventilator other than window.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:17Z
Check the source to verify that this is the current version before use.

3.10.20 Rearview mirrors. Outside rearview mirrors, each with at least three supporting arms, shall be mounted on each side of the cab. Combination have at least 50 square inches of flat reflective area and a convex surface having at least 20 square inches of reflective area.

3.10.21 Horn. Manufacturer's standard electric horn shall be furnished. When specified (see 6.2), in addition, an air operated horn shall be furnished.

3.10.22 Engine hour meter. An engine hour meter having a totalizing mechanism of not less than 9,999 hours shall be furnished for the chassis engine to register accurately the number of hours of operating time. The meter shall be of rugged construction to insure continuous trouble free performance under severe operating conditions. Engine hour meter shall be mounted on the cab instrument panel or in the engine compartment in a readable location.

3.10.23 Heavy-duty cooling system. A heavy-duty cooling system shall be furnished which will maintain coolant at a temperature below the boiling point with the truck loaded to rated gvw and operated at 10,000 feet above sea level or in an ambient air temperature of not less than 125°F. Coolant system shall include a de-aeration system, a surge tank or a coolant recovery reservoir of not less than 2 quart capacity. A low coolant level or high coolant temperature alarm buzzer and red indicator warning light shall be provided on the dash instrument panel.

3.11 Concrete placing equipment. The concrete pump, when furnished (see 3.14), and concrete placing equipment shall be truck mounted. The mounting platform shall consist of a welded, riveted, bolted, (or combination thereof) fabricated structural steel frame, with necessary braces, reinforcements, and brackets to provide rigid support for the mounted equipment under all operating conditions and at maximum loads. The platform shall be integral with the outrigger support frame (see 3.15.1). Loads to the platform resulting from elevating and slewing the boom shall not be transmitted to the truck chassis when stabilized by outriggers.

3.12 Rotation assembly. The rotation assembly shall be slewed hydraulically. The rotation assembly shall be capable of slewing in either direction in the horizontal plane. The rotation assembly shall slew the extended boom through an arc of not less than 315°. Rotation assembly device shall be capable of remaining in a fixed position when the hydraulics are de-energized. The rotation assembly shall be capable of smoothly slewing the boom assembly up and down a 5 percent grade when loaded to rated working capacity, and producing the maximum attainable tipping moment. With the unit on a 5 percent grade, the rotation assembly shall prevent boom movement (rotation) of more than 1 inch per hour when measured at the boom tip with hydraulic fluid stabilized at ambient air temperature and the hydraulics de-energized.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:17Z
Check the source to verify that this is the current version before use.

3.13 Boom assembly. The boom assembly shall be a three section, articulated (folding-arm) type and shall be hydraulically operated. The boom assembly shall be attached to the rotation assembly utilizing self-alining bearings. Under dynamic conditions, the boom assembly shall support the maximum working load placed on the boom tip in any extended position without permanent deformation of any component in or along the boom or its supporting structure. With the unit on a 5 percent grade, the elevating mechanism shall prevent the extended boom from vertical settling more than 2 inches per hour. Settling shall be measured at the front of the boom tip with hydraulic fluid stabilized at ambient air temperature and the hydraulics de-energized.

3.13.1 Vertical reach. The boom cylinders shall be capable of articulating the boom sections, with the normal working load, to a minumum heigh of 60 feet. The height shall be measured from ground level.

3.13.2 Horizontal reach. With the boom in its most horizontal reach of not less than 50 feet. The reach shall be measured from the boom tip to the axis of the turntable.

3.13.3 Boom constructions. The boom sections shall be designed, engineered, and constructed of materials that display the maximum weight conducted to the configuration and maximum performance requirements specified herein. Hydraulic lines and the associated parts which are fastened to or inside the boom sections shall be positioned for easy inspection, ready maintenance or possible replacement, and shall have adequate protection against damage afforded by the configuration of the sections. Bearing surfaces of the boom at the rotation assembly connection and at the boom section joints which are subjected to continuous wear shall be provided with replaceable antifriction bearings or bushings. Lubrication fittings shall be provided for all surfaces requiring lubrication. Bearing surfaces shall be adequate to resist torsional movement when inherent with the design.

3.13.4 Delivery system. The delivery system shall consist of an all steel pipeline having a minimum nominal diameter of not less than 4 inches, steel swivel joints at boom connections, and a hose support assembly complete with hose shall be provided. The pipeline shall not be an integral part of the boom. The pipes and joints shall be readily replaceable.

3.13.5 Boom operation. The rotation assembly and the boom sections shall be hydraulically actuated by separate mechanism and shall be capable of operating independently and in any combination of raising, lowering, and slewing simultaneously. When in combination, hand metering of the controls may be necessary. Boom articulation and rotation shall be smooth with no lurching or binding.

3.13.6 Controls. A complete set of hand-operated controls shall be provided to actuate the rotation assembly, boom section cylinders, and outriggers. In addition, when a concrete pump assembly is furnished (see 3.14), hand operated controls to actuate and regulate pump assembly functions

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:17Z
Check the source to verify that this is the current version before use.

shall be provided. Controls shall be self-centering and shall revert to a neutral position when released by the operator. Placement of controls shall be so as to prevent inadvertent actuation by the operator.

3.13.6.1 Remote electromechanical control. A remote electromechanical control shall be provided to slew the turntable, to articulate the boom, and to adjust hydraulic pump speed. Control cable length shall not be less than 100 feet. In addition, when a concrete pump is furnished, controls to regulate pump functions shall be provided.

3.13.7 Boom storage. For transport the boom sections shall fold to essentially horizontal positions in the vertical plane parallel to the centering of the truck. A boom support saddle, if necessary to prevent damage during transport, shall be provided.

3.14 Concrete pump assembly. When specified (see 6.2), a concrete pump assembly shall be provided. The concrete pump assembly shall consist of the concrete pump and loading hooper, with supports and accessories as required, mounted on the truck and connected to perform the required function. The concrete pump shall be of twin cylinder, piston displacement type. The pump shall develop material delivery pressures up to 800 pound-force per square inch (psi). The surfaces of the material regulating suction and discharge valve cylinders, pistons, and piston rods shall be chromium plated not less than 0.005 inch in thickness or case hardening material shall be used or in combination. A means shall be provided to regulate the discharge of the concrete pump.

3.14.1 Performance requirements. The pump unit shall perform the following functions:

a. Deliver a minumum output capacity of 70 cubic yards per hour through the boom delivery system. The concrete mix shall be of the low slump, high quality type, consisting of 1-1/2 inch minus aggregates.
b. Concrete mixes discharge from the pump shall form a smooth laminar flow.
c. Develop up to 800 psi pulsating line pressure without stagnation and segregation of the concrete mixes.

3.14.2 Cleaning system. A cleaning system which will enable a trained crew to remove residual concrete deposits from internal surfaces of the loading hopper, pump cylinders and delivery system is not more than 10 minutes shall be provided. The cleaning system shall use water pump and air system from the feed water system (see 3.14.5). The cleaning system shall include a hose, not less than 25 feet in length, complete with fittings and nozzle, for cleaning and washdown of external areas of the concrete pump and associated equipment. The hose shall be suitable for the intended use and water pressure. A hose storage rack shall be furnished and mounted on the unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:17Z
Check the source to verify that this is the current version before use.

3.14.3 Drive system. The concrete pump and all hydraulically operated equipment shall be powered by the truck pto or by and auxiliary engine of a commercial design, that has proven satisfactory in extensive use, and for which repair parts are readily obtainable. Auxiliary engine shall produce the required pump performance on diesel fuel. If an auxiliary engine is furnished, an engine pto may be utilized to drive the hydraulic pump(s) and other components. The power and speed of the engine shall meet the performance requirements as specified. The auxiliary engine shall be furnished as a minimum the accessories as follows:

a. An instrument panel mounted at operator's station, complete with a lubricating oil pressure gage; a fuel oil pressure gage which indicates low, normal, and excessive pressures; a cooling liquid-temperature indicator; a battery charging indicator; a tachometer; and an hour meter.
b. A fuel tank with sufficient capacity for 8 hours normal operation.
c. A 12-V electric cranking system.
d. Battery(s) to be moisture sealed and dry charged.
e. A charging alternator with a rating of not less than 40A.
f. A cooling-liquid high temperature safety device.
g. A lubricating oil safety device.
h. An air cleaner.
i. A torque converter, if furnished, oil temperature indicator.
j. An engine housing.

3.14.4 Loading hopper. Loading hopper shall be fabricated of heavy gage, corrosion- and abrasion-resistant sheet steel reinforced to support a load of not less than 12 cubic feet of concrete mix without permanent distortion. Loading hopper receiving opening shall be equipped with a heavy-duty steel screen of not less than 1-inch square openings, and shall have sufficient slope to permit the heavy concrete mixes to fall into the cylinders without blockage. A power riven agitator, either of paddle or spiral auger type, shall be provided the full length of loading hopper. The agitator shall be fabricated of steel with corrosion- and wear-resistant qualities. When paddle type agitator is furnished, paddle tips shall be replaceable.

3.14.5 Feed water system. A feed water system, icluding a water storage tank, water air system pump, and piping system, as required, to provide a complete feed water system shall be provided.

3.14.5.1 Water storage tank. The water storage tank shall have a capacity of not less than 80 gallons. A handhole, shall be provided on the tank top filling, and a provision on the tank bottom for draining.

3.15 Stabilizing equipment.

3.15.1 Outriggers. Not less than two hydraulically operated outriggers shall be provided. Outriggers shall enable leveling the unit on inclined surfaces and shall extend to insure adequate stability with relation to the ground, while operating the boom in all positions. The outrigger support

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:17Z
Check the source to verify that this is the current version before use.

frame shall be integrally constructed with the rotation assembly mounting platform. All loadings from the boom shall be transmitted through the outrigger support frame. Outrigger cylinder hydraulic fittings and hoses shall be readily accessible for service. Outriggers shall be designed so that they may be extended over curbings and sidewalks when operating in narrow areas.

3.15.2 Outrigger controls. An individual control for each outrigger shall be furnished. A check valve shall be provided for each outrigger to block the flow of hydraulic fluid to prevent movement of the outrigger in the event of loss of hydraulic pressure. All outrigger controls shall be installed in a readily accessible location which permits a check valve which holds outrigger in the retracted position without settling.

3.16 Hydraulic system.

3.16.1 Hydraulic fluid reservoir. A hydraulic fluid reservoir shall be provided and shall ahve sufficient capacity to hold the required volume of hydraulic fluid, with not less than 10 percent for expansion space. The reservoir shall be provided with a filler opening, a removable strainer, and a filler cap. Vented filler caps shall have means to preclude fluid splashing out of the reservoir during movement of vehicle. Provisions shall be made for draining of the reservoir. The reservoir drain shall be so located that the fluid will not drain on other components of the unit. A sight level or dip stick gage shall be furnished to determine level of fluid.

3.16.2 Hydraulic oil filter. Manufacturer's standard hydraulic oil filtration system shall be furnished.

3.16.3 Hydraulic pump system. A hydraulic pump with necessary pressure and flow capacities to operate all hydraulically powered components shall be provided. The hydraulic system shall include and adjustable pressure relief valve or unloading valve with the range of adjustment necessary to operate all hydraulically powered components.

3.16.4 Hydraulic cylinders. All hydraulic cylinders shall incorporate seals to prevent entrance of moisture and dirt into the cylinders and leaks of hydraulic fluid from the cylinders. All hydraulic cylinders for actuating the boom shall be provided with pilot check valves to prevent descent of the boom assembly in the event of loss of hydraulic pressure.

3.16.5 Hydraulic lines and fittings. Hydraulic lines shall be located to facilitate maximum ease of removal and installation, or visual inspection for failure, and maximum protection from damage during operation of the boom assembly. Flexible hydraulic hose or collector rings shall be used where flexing of hydraulic lines is encountered, and where installation of components requires a flexible connection. Hydraulic lines and fittings shall be installed in such a manner so as to preclude vibration and chafing. Hose assemblies and all seamless steel hydraulic lines shall be of adequate size for rate of slow, temperature, and pressures required to operate the system smoothly and efficiently.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:17Z
Check the source to verify that this is the current version before use.

3.17 Lubrication. Means for lubrication shall be in accordance with the manufacturer's standard practice. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

3.18 Instruction plates. The unit shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the unit; and all inherent dangers which can stem from unsafe operations or an unsafe operating position. Where applicable the words "Warning" or "Caution" shall appear and shall be red. The shade of red shall conform to FED-STD-595 No. 31136. Plates shall be of a material which will last and remain legible for the life of the equipment.

3.19 Identification plate. An identification plate will be furnished by the contracting officer for each unit. The contractor shall stamp all necessary data in the blank spaces of the plate provided for that purpose, and securely affix a plate to each unit in a conspicuous place with nonferrous screws, rivets, or bolts not less than 1/8-inch in diameter. The applicable nomenclature contained in the contract item description shall be placed in the top blank.

3.20 Identification marking. The truck shall be marked for identification in accordance with MIL-STD-1223.

3.21 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.22 Workmanship.

3.22.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:17Z
Check the source to verify that this is the current version before use.

3.22.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.22.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.22.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.22.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:17Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. First article inspection shall be performed on one complete unit when a first article sample is required (see 3.2 and 6.4). This inspection shall include the examination of 4.3 and the tests of 4.4 through 4.4.1.5. If first article sample is furnished with a concrete pump assembly the tests of 4.4.1.6 and 4.4.1.7 shall be included in the inspection. The first article may be a standard production item from the contractor's current inventory, provided the unit meets the requirements of this specification and is representative of the design, construction, and manufacturing technique applicable to the remaining units to be furnised under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the production test of 4.4.2, and the packaging inspection of 4.5.

4.3 Examination. Each unit shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. The first article shall be tested in accordance with 4.2.1. Production units shall be tested in accordance with 4.2.2.

4.4.1 First article tests. The unit shall be fueled and serviced in accordance with the manufacturer's recommendations preparatory to testing. The first article shall be tested to verify compliance with the performance requirements of this specification, as follows:

4.4.1.1. Outriggers. Outriggers shall be extended/lowered to stabilize the unit. With the power of the hydraulic system stopped, each outrigger control shall be actuated to determine if the holding valves maintain the static position of the appropriate outrigger. Outrigger piston rod shall be observed to determine if internal or external leakage or rod movement has occured.

4.4.1.2 Rotation assembly. With the unit stabilized on a 5 percent side grade the extended boom shall be raised to approximately a 45° angle, longitudial to the centerline of the truck. Horizontal creep shall be measured in accordance with 3.12. The rotation assembly shall be slewed, with an elevated boom, through an arc of not less than 315° to verify conformance with 3.12.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:17Z
Check the source to verify that this is the current version before use.

4.4.1.3 Boom. With the unit stabilized by the outriggers, the boom shall be raised and articulated to the maximum vertical height and lowered to the most horizontal position to verify height and length requirements of 3.13.1 and 3.13.2. The boom shall operate in all positions without evidence of binding or lurching. The boom shall be elevated in an extended position to approximately a 45° angle, the boom tip height shall be measured to verify that settling does not exceed the requirements of 3.13. With the power to the hydraulic system stopped, each boom section cylinder control valve shall be actuated to determine if the holding valves maintain the static position of the appropriate boom section.

4.4.1.4 Remote electromechanical control. Boom articulation, slewing, and hydraulic pump speed, and when furnished concrete pump functions, shall be controlled by hand-operated controls (see 3.13.6), and by using the remote electromechanical control (see 3.13.6.1), attached to a suitable cable not less than 100 feet in length.

4.4.1.5 Road test. The unit shall be road tested at a minimum of 20 miles to verify that the unit will operate in accordance with 3.7 and contractual requirements.

4.4.1.6 Pump and system test. Raise and articulate the boom to a height of 50 feet and a reach of not less than 50 feet (from center of rotation to point of perpendicular drop of delivery system discharge pipe or hose). Charge 10 cubic yards of freshly prepared 2-inch slump concrete, consisting of 1-1/2 inch minus aggregate into the loading hopper and pump the concrete with periodical manipulations of the remote electromechanical control to verify proper functions and proper operational control for concrete placement. Control manipulation shall include the following: Slew the boom left and right at least twice, elevate the boom and articulate each section in various degress of angle until the maximum height is reached. Failure to discharge concrete at various boom positions or operations not in accordance with the specified requirements in section 3 shall be cause for rejection.

4.4.1.7 Flushing and cleaning test. No earlier than 30 minutes after the tests in 4.4.1.6 are completed, the entire pumping and delivery system shall undergo a flushing and cleaning test, using the materials and procedures in accordance with the contractor's recommended practice to determine conformance to 3.14.2. Disassemble all the delivery system piping after the test, and observe the degree of cleanliness in the pipe, pump(s), and the loading hopper.

4.4.2 Production test. Each unit shall be fueled and serviced in accordance with the manufacturer's recommendations preparatory to testing. Hand operated and remote electromechanical controls shall be actuated to verify correct functioning of equipment without concrete.

4.5 Packaging inspection. The preservation packing and marking of the item shall be inspected to verify conformance to the requirements of section 5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:17Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Commercial. The complete equipment shall be preserved and packed in accordance with the contractor's standard practice in a manner to prevent deterioration and damage during shipment. Preparation for delivery shall comply with applicable carrier rules and regulations.

6. NOTES

6.1 Intended use. The unit is intended for conveying pumped concrete above and below grade and over typical job site obstructions.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. When specifications or standards shall be other than as specified (see 2.1.1).
c. When first article is required for inspection and approval (see 3.2 and 6.4).
d. When dry-charged type batteries are required (see 3.10.2.4).
e. When automatic or semiautomatic transmission is required (see 3.10.5).
f. When an air horn is required (see 3.10.21).
g. When a concrete pump assembly is required (see 3.14).
h. Color of finish coat required (see 3.21).
i When other than commercial packaging and packing is required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data shall be delivered by the contractor in accordance with the contract or purchase order requirements.

6.4 First article. When a first article inspection is required (see 6.2), the item will be tested and should be a first production item or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:17Z
Check the source to verify that this is the current version before use.

6.5 Subject term (key word) listing

Diesel engine.
Placing.
Pumped concrete.
Safety.
Truck.

Custodian:
Navy - YD

Preparing activity:
Navy - YD

Project 3895-N249

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T22:17Z
Check the source to verify that this is the current version before use.